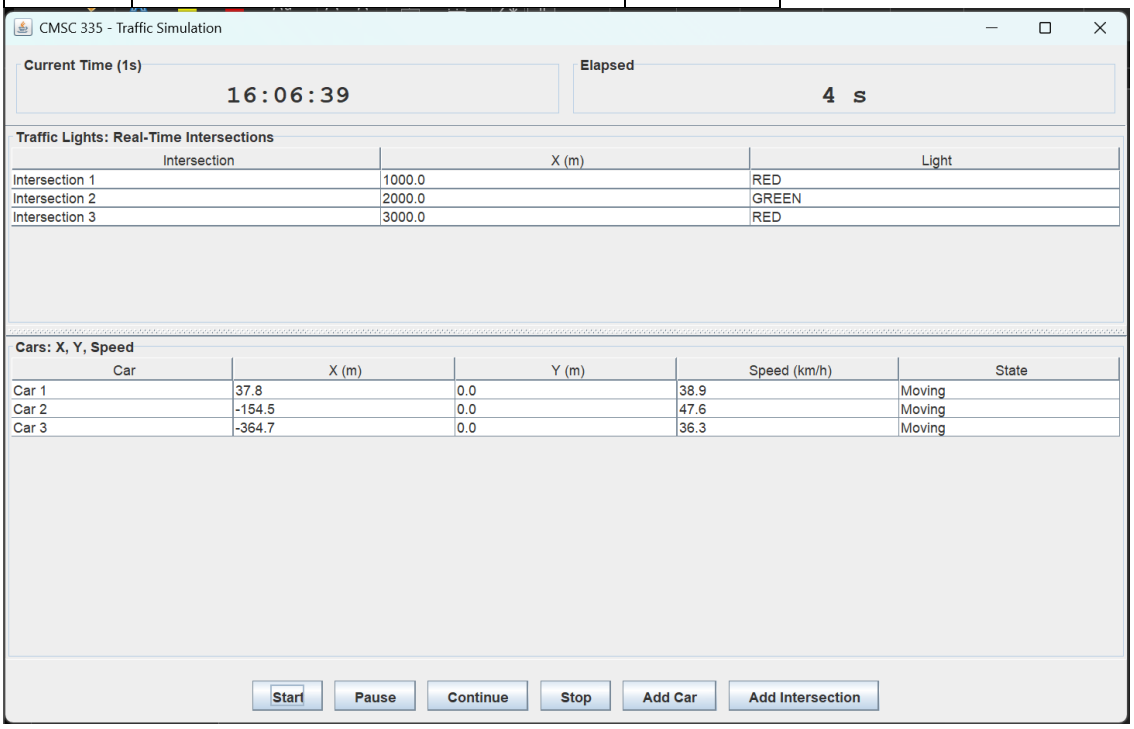
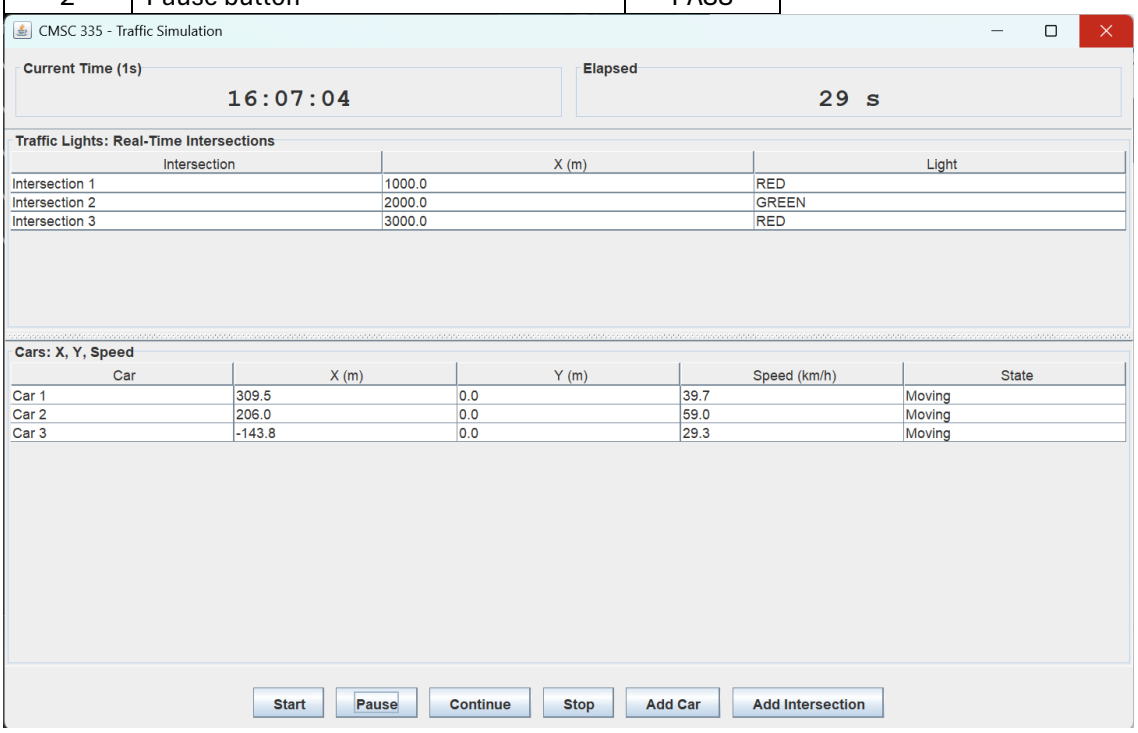
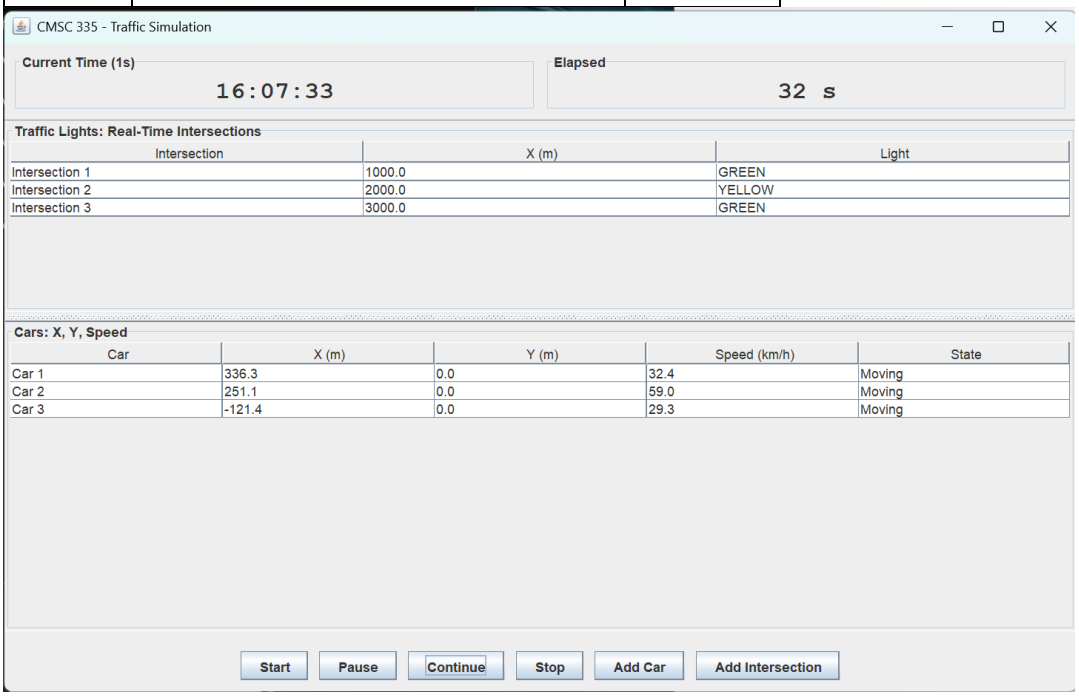
| Test # | Description | PASS/FAIL |
|---|---|---|
| 1 | Start button | PASS |



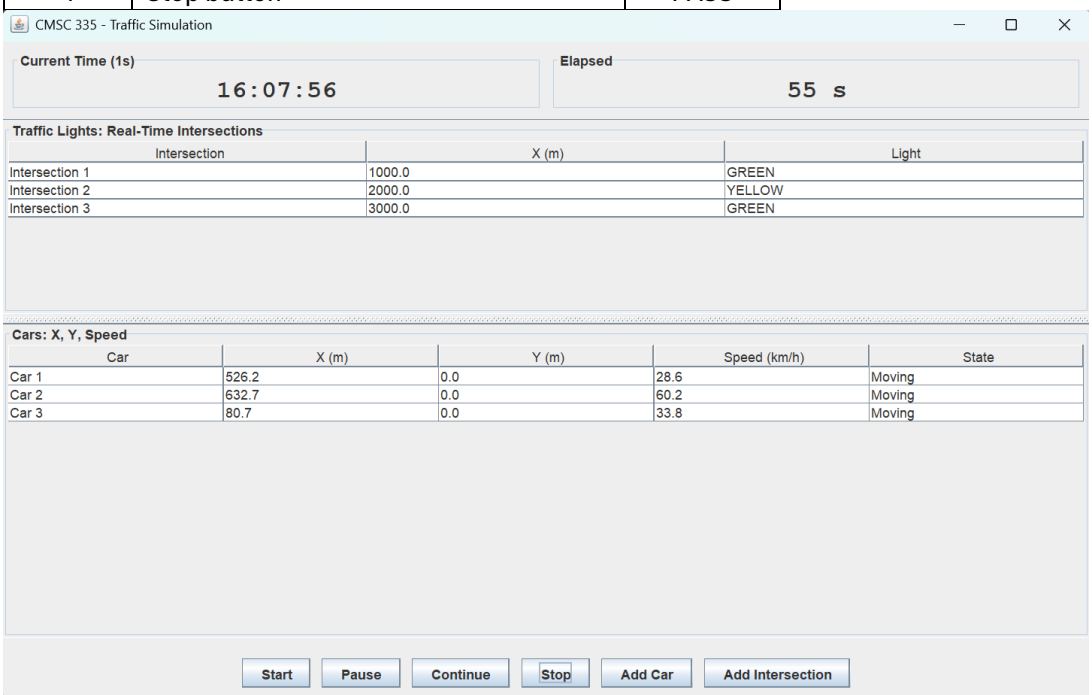

| Test # | Description | PASS/FAIL |
|---|---|---|
| 2 | Pause button | PASS |



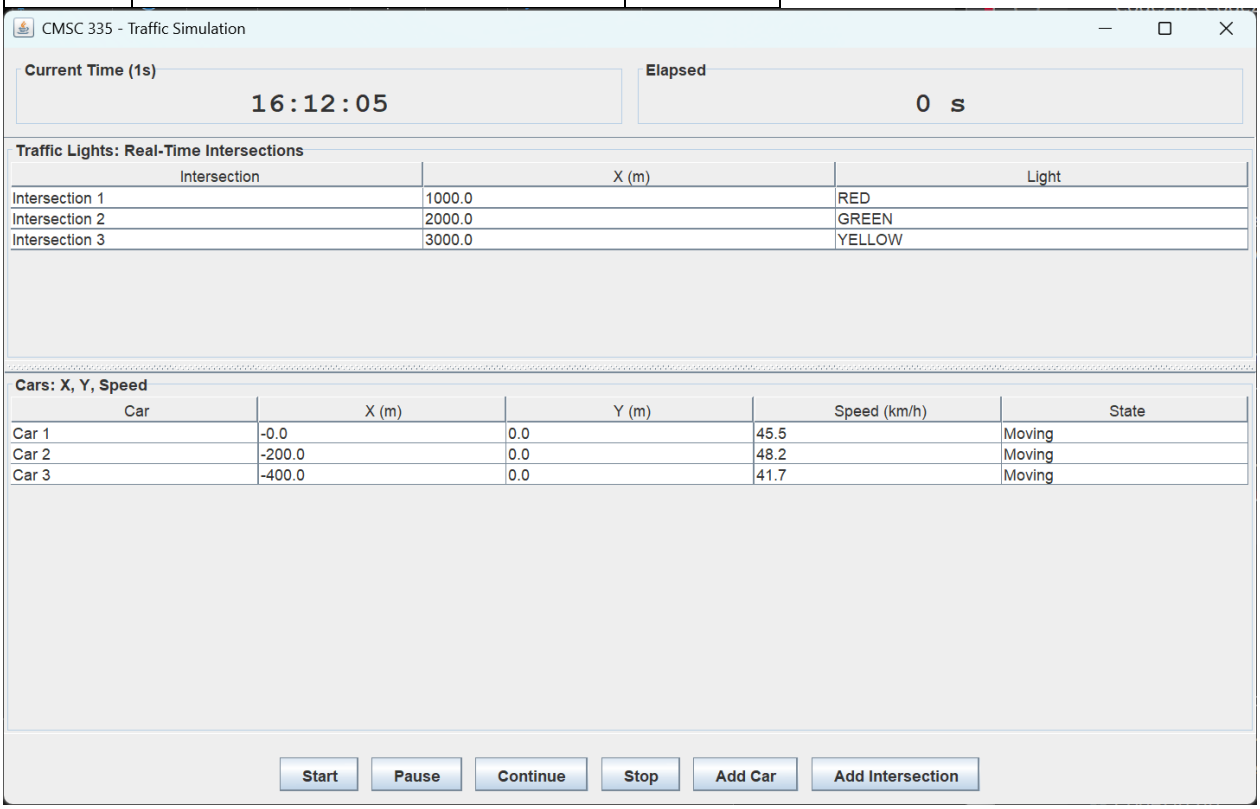
| Test # | Description | PASS/FAIL |
|---|---|---|
| 3 | Continue button | PASS |



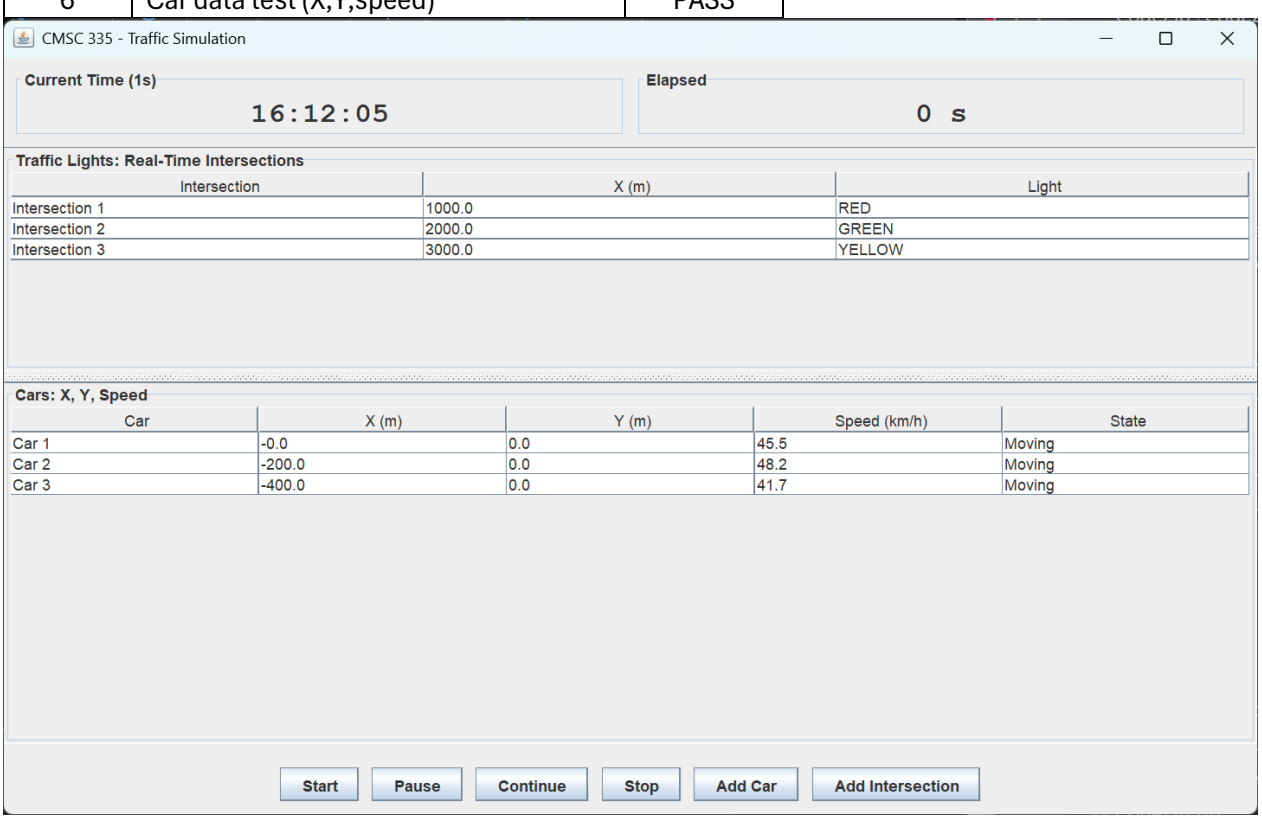

| Test # | Description | PASS/FAIL |
|---|---|---|
| 4 | Stop button | PASS |



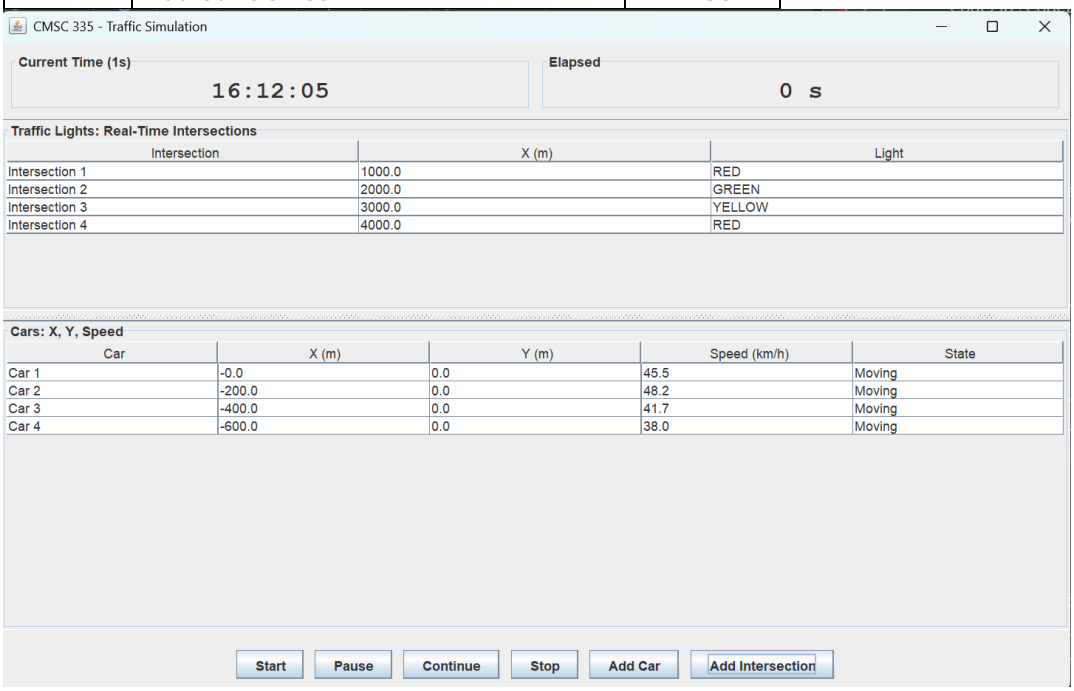
| Test # | Description | PASS/FAIL |
|---|---|---|
| 5 | Three intersections | PASS |



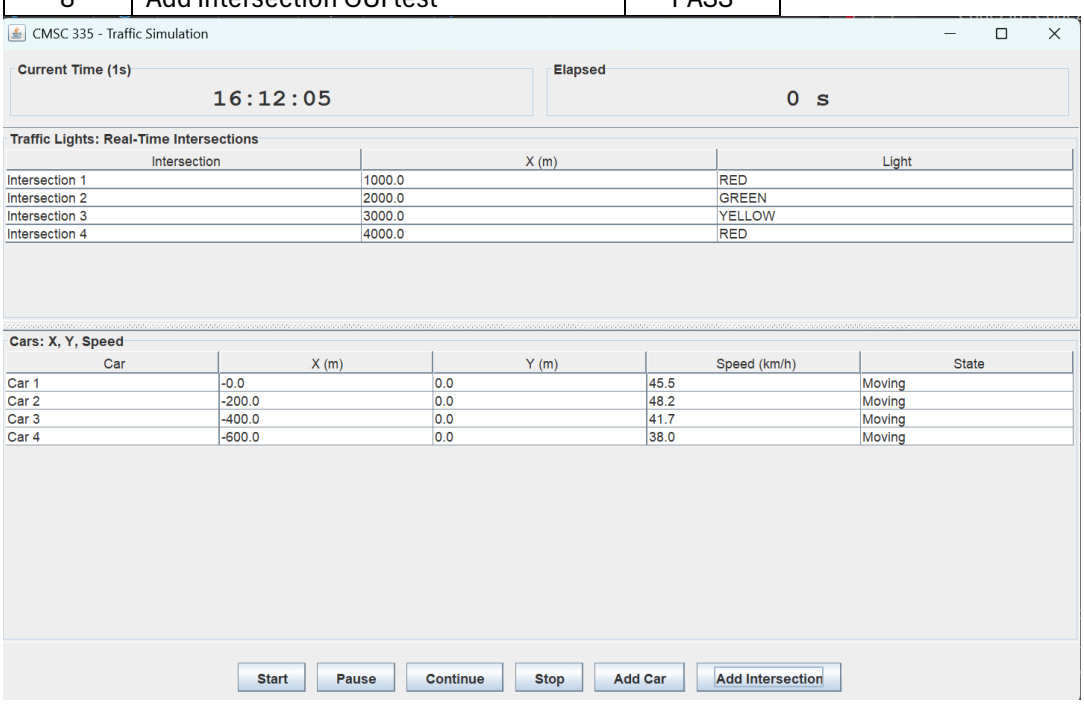

| Test # | Description | PASS/FAIL |
|---|---|---|
| 6 | Car data test (X,Y,speed) | PASS |



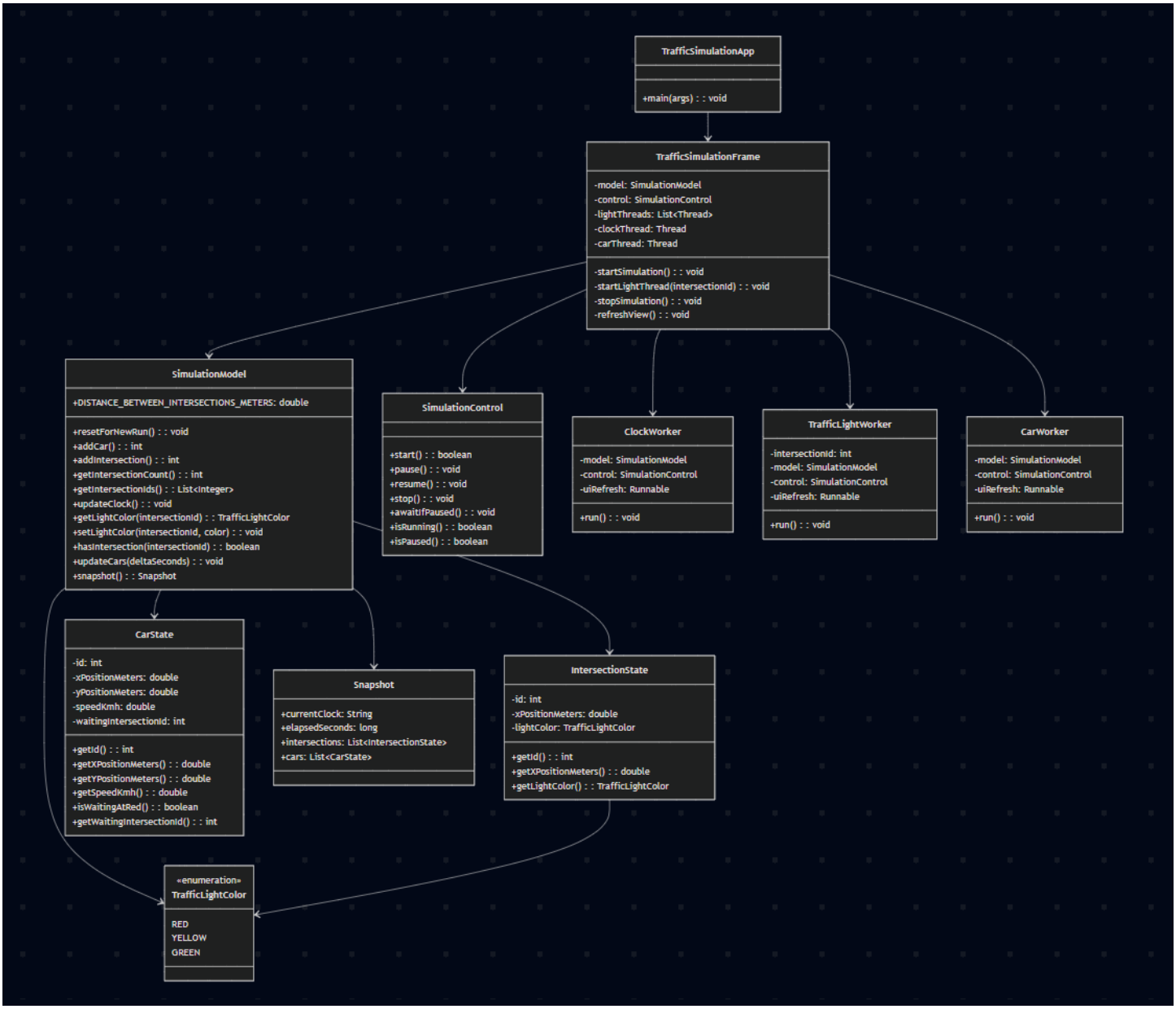
| Test # | Description | PASS/FAIL |
|---|---|---|
| 7 | Add Car GUI test | PASS |




| Test # | Description | PASS/FAIL |
|---|---|---|
| 8 | Add Intersection GUI test | PASS |



# UML Diagram


TrafficSimulationApp
+main(args) : : void
TrafficSimulationFrame
-model: SimulationModel
-control: SimulationControl
-lightThreads: List<Thread>
-clockThread: Thread
-carThread: Thread
-startSimulation() : : void
-startLightThread(intersectionId) : : void
-stopSimulation() : : void
-refreshView() : : void
SimulationModel
+DISTANCE_BETWEEN_INTERSECTIONS_METERS: double
+resetForNewRun() : : void
+addCar() : : int
+addIntersection() : : int
+getIntersectionCount() : : int
+getIntersectionIds() : : List<Integer>
+updateClock() : : void
+getLightColor(intersectionId) : : TrafficLightColor
+setLightColor(intersectionId, color) : : void
+hasIntersection(intersectionId) : : boolean
+updateCars(deltaSeconds) : : void
+snapshot() : : Snapshot
SimulationControl
+start() : : boolean
+pause() : : void
+resume() : : void
+stop() : : void
+awaitIfPaused() : : void
+isRunning() : : boolean
+isPaused() : : boolean
ClockWorker
-model: SimulationModel
-control: SimulationControl
-uiRefresh: Runnable
+run() : : void
TrafficLightWorker
-intersectionId: int
-model: SimulationModel
-control: SimulationControl
-uiRefresh: Runnable
+run() : : void
CarWorker
-model: SimulationModel
-control: SimulationControl
-uiRefresh: Runnable
+run() : : void
CarState
-id: int
-xPositionMeters: double
-yPositionMeters: double
-speedKmh: double
-waitingIntersectionId: int
+getId() : : int
+getXPositionMeters() : : double
+getYPositionMeters() : : double
+getSpeedKmh() : : double
+isWaitingAtRed() : : boolean
+getWaitingIntersectionId() : : int
Snapshot
+currentClock: String
+elapsedSeconds: long
+intersections: List<IntersectionState>
+cars: List<CarState>
IntersectionState
-id: int
-xPositionMeters: double
-lightColor: TrafficLightColor
+getId() : : int
+getXPositionMeters() : : double
+getLightColor() : : TrafficLightColor
«enumeration»
TrafficLightColor
RED
YELLOW
GREEN